An Improved Lower Bound for the $3x + 1$ Problem: $\pi_a(x) \geq x^{0.895}$

M. Sharpe*

June 12, 2026; revised August 29, 2026

Abstract

For $a \not\equiv 0 \pmod 3$ let $\pi_a(x)$ count the integers $n \leq x$ whose forward orbit under the $3x + 1$ function contains a. Krasikov and Lagarias (2003) proved $\pi_a(x) \geq x^{0.84}$ for all sufficiently large x, by exhibiting a feasible solution to an explicit linear program $L_k^{NT}(\lambda)$ over the 3^{k-1} congruence classes $m \equiv 2 \pmod 3$ of modulus 3^k, at level $k = 11$; this has remained the best published exponent. We improve the bound to

$$\pi_a(x) \;\geq\; x^{179/200} \;=\; x^{0.895} \qquad \text{for all sufficiently large } x,$$

using their theorem verbatim and new feasible solutions at level $k = 17$ (43,046,721 classes). The computational unlock is structural: for fixed λ the program is a feasibility question $c \leq F_\lambda(c)$ for a monotone, homogeneous, concave map, so a feasible point is found by nonlinear power iteration (Collatz–Wielandt) rather than linear programming, in time linear in the number of classes per sweep. Soundness does not rest on floating point: for each claimed exponent p/q we exhibit an integer vector and verify every inequality of $L_k^{NT}(2^{p/q})$ in exact integer arithmetic, with the transcendental coefficients replaced by rational lower bounds certified via integer power comparisons. The method reproduces the complete published table of Krasikov–Lagarias ($2 \leq k \leq 11$) to all seven printed decimals, and yields certified exponents $0.853, 0.863, 0.8724, 0.8812, 0.888, 0.895$ at $k = 12, \ldots, 17$. Separately, the record itself is now MACHINE-VERIFIED: a Lean 4 formalization of the Krasikov–Lagarias inequality system — including its "advanced" term, which their proof handles by an elimination argument and which we handle by a strong induction over roots on the measure $10t + \lfloor \log_2 a^{498} \rfloor$ — together with a 177,147-class certificate decided by the Lean kernel, gives $\pi_1(x) = \Omega(x^\gamma)$ with $\gamma = 50 \log_2(101182/100000) = 0.8476\ldots > 0.84$ at level $k = 12$ and $\gamma = 0.8569\ldots$ at $k = 13$, the first kernel-checked improvements of the 2003 bound (Section 6).

1 Introduction

The $3x + 1$ function is $T(n) = n/2$ for even n and $T(n) = (3n + 1)/2$ for odd n; the $3x + 1$ conjecture asserts that every $n \geq 1$ eventually reaches 1 under iteration [6, 7]. For a positive integer $a \not\equiv 0 \pmod 3$, let

$$\pi_a(x) \;:=\; \#\{\, 1 \leq n \leq x \;:\; T^{(j)}(n) = a \text{ for some } j \geq 0 \,\}.$$

*msharpe@irendi.com. The computations, the certification scheme, and this text were produced in collaboration with Claude (Anthropic). All final verifications are exact integer computations; see Section 4.

A sequence of results, beginning with Crandall [2] in 1978, established lower bounds $\pi_1(x) > x^\beta$ for increasing exponents β: the difference-inequality method introduced by Krasikov [4] gave $x^{0.43}$, Wirsching [10] $x^{0.48}$, the nonlinear-programming approach of Applegate and Lagarias [1] $x^{0.81}$, and Krasikov–Lagarias [5] reached

$$\pi_a(x) \;\geq\; x^{0.84} \qquad (a \not\equiv 0 \bmod 3,\; x \geq x_0(a)),$$

by a computation, performed in 2002 at level $k = 11$, on a linear program family $L_k^{NT}(\lambda)$ recalled below. This exponent has remained the best published bound: the 2021 overview [7] cites it as the state of the art, and we are not aware of any later improvement. (Tao's celebrated result that almost all orbits attain almost bounded values [9] concerns logarithmic density of a different event and neither implies nor is implied by pointwise bounds on π_a.)

Our result is:

Theorem 1.1. *For every positive integer $a \not\equiv 0 \pmod 3$ there is $x_0(a)$ such that*

$$\pi_a(x) \;\geq\; x^{179/200} \;=\; x^{0.895} \qquad \textit{for all } x \geq x_0(a).$$

The theorem follows from the Krasikov–Lagarias framework *verbatim* — their Theorem 2.2 converts any feasible solution of $L_k^{NT}(\lambda)$ into the bound with exponent $\gamma = \log_2 \lambda$, and they prove feasibility lifts from k to $k+1$, so larger k never hurts — together with a new feasible solution at $k = 17$, over 43,046,721 congruence classes, with $\lambda = 2^{179/200}$. The contribution of this paper is therefore computational, in two parts that we believe are of independent methodological interest:

1. **Feasibility without linear programming (Section 3).** For fixed λ the program $L_k^{NT}(\lambda)$ asks for a positive bounded vector c with $c \leq F_\lambda(c)$, where F_λ is monotone, positively homogeneous, and concave (each coordinate is a linear term plus a minimum of three coordinates, with positive coefficients). Any vector satisfying $\min_m F_\lambda(c)_m / c_m \geq 1$ is itself a feasible point — this is the easy direction of the Collatz–Wielandt characterization for monotone homogeneous maps (the coincidence of names is pleasant but accidental: Lothar Collatz of the eigenvalue bound is Lothar Collatz of the conjecture). Normalized power iteration on F_λ produces such vectors whenever λ is below the feasibility supremum λ_k. Each sweep costs $O(3^k)$, which is what moves the frontier from 3^{10} classes (LP, 2002 hardware) to 3^{16} and beyond.

2. **Exact certification (Section 4).** Floating point is used only to *find* candidate vectors. Each claimed exponent is a rational p/q; the three coefficients of $L_k^{NT}(2^{p/q})$ are bounded below by dyadic rationals whose correctness is an integer power comparison; and the candidate vector, scaled and floored to integers, is checked against every inequality in exact integer arithmetic — all 43,046,721 of them at $k = 17$. Since the coefficients enter the right-hand sides positively, rounding them down only strengthens the verified system; soundness is one-directional throughout.

As validation, the same machinery reproduces every entry of the published Krasikov–Lagarias table ($2 \leq k \leq 11$) to all seven printed decimal places, including their record point $\lambda_{11} = 1.7922310$ (Section 5).

2 The Krasikov–Lagarias program

We recall the objects of [5], to which we refer for proofs. Throughout $\alpha := \log_2 3 = 1.58496\ldots$ and

$$[3^k] \;:=\; \{\, m \bmod 3^k \;:\; m \equiv 2 \pmod 3 \,\}, \qquad |[3^k]| = 3^{k-1}.$$

For $m \in [3^k]$ define $\phi_k^m(y) := \inf\{\pi_a^*(2^y a) : a \equiv m \bmod 3^k,\ a \text{ not in a cycle}\}$, where π_a^* is the variant of π_a counting orbits that stay $\leq x$ (see [5, §2]). These functions satisfy a system of difference inequalities $\mathcal{I}_k$ [5, Prop. 2.1], and Krasikov–Lagarias associate to it the linear program family

$$L_k^{NT}(\lambda): \quad \text{minimize } C_k^{\max} \text{ subject to}$$

$$\begin{array}{lll}
\text{(L0)} & 1 \leq c_k^m \leq C_k^{\max} & m \in [3^k], \\
\text{(L1)} & c_k^m \leq c_k^{4m}\lambda^{-2} + c_{k-1}^{(4m-2)/3}\lambda^{\alpha-2} & m \equiv 2 \pmod 9, \\
\text{(L2)} & c_k^m \leq c_k^{4m}\lambda^{-2} & m \equiv 5 \pmod 9, \\
\text{(L3)} & c_k^m \leq c_k^{4m}\lambda^{-2} + c_{k-1}^{(2m-1)/3}\lambda^{\alpha-1} & m \equiv 8 \pmod 9, \\
\text{(L4)} & c_{k-1}^u \leq c_k^u,\ \ c_{k-1}^u \leq c_k^{u+3^{k-1}},\ \ c_{k-1}^u \leq c_k^{u+2\cdot 3^{k-1}} & u \in [3^{k-1}],
\end{array}$$

with indices taken modulo 3^k (note $4m$, $(4m-2)/3$ and $(2m-1)/3$ remain $\equiv 2 \bmod 3$). Their main results, which we use as black boxes:

Theorem 2.1 ([5, Thm. 2.2 and §6])**.** *If $1 \leq \lambda \leq 2$ and $L_k^{NT}(\lambda)$ has a feasible solution with principal variables $\{c_k^m\}$, then $\phi_k^m(y) \geq \Delta_1 c_k^m \lambda^y$ for all $y \geq 0$, with $\Delta_1 = (4 \max_m c_k^m)^{-1}$; consequently, for every $a \not\equiv 0 \pmod 3$, $\pi_a(x) \geq x^{\log_2 \lambda}$ for all sufficiently large x.*

Proposition 2.2 ([5, §6])**.** *If $L_k^{NT}(\lambda)$ is feasible, so is $L_{k+1}^{NT}(\lambda)$ (set $c_{k+1}^{m+j\cdot 3^k} := c_k^m$ for $j = 0, 1, 2$). Hence $\lambda_k := \sup\{\lambda : L_k^{NT}(\lambda)$ feasible$\}$ is non-decreasing in k.*

It is an open problem whether $\lambda_k \to 2$, which would give $\pi_a(x) \geq x^{1-\epsilon}$ for every $\epsilon > 0$ [5, §6].

3 Feasibility as a nonlinear eigenproblem

Fix λ and eliminate the auxiliary variables: given any principal vector $c = (c_k^m)_{m \in [3^k]}$, the optimal choice in (L4) is $\bar{c}_{k-1}^u := \min\{c_k^u,\, c_k^{u+3^{k-1}},\, c_k^{u+2\cdot 3^{k-1}}\}$ [5, eq. (2.15)]. Substituting into (L1)–(L3) defines the map $F_\lambda : \mathbb{R}_{>0}^{[3^k]} \to \mathbb{R}_{>0}^{[3^k]}$,

$$F_\lambda(c)_m \;=\; \begin{cases} \lambda^{-2} c_{4m} + \lambda^{\alpha-2}\, \bar{c}_{(4m-2)/3} & m \equiv 2 \pmod 9, \\ \lambda^{-2} c_{4m} & m \equiv 5 \pmod 9, \\ \lambda^{-2} c_{4m} + \lambda^{\alpha-1}\, \bar{c}_{(2m-1)/3} & m \equiv 8 \pmod 9, \end{cases}$$

and $L_k^{NT}(\lambda)$ is feasible if and only if there is a vector c with $1 \leq c_m$, bounded, and $c \leq F_\lambda(c)$ pointwise (then $C^{\max} := \max c$). The map F_λ is monotone ($c \leq c' \Rightarrow F(c) \leq F(c')$), positively homogeneous ($F(tc) = tF(c)$ for $t > 0$), and concave; all coefficients are positive.

Lemma 3.1. *If $c > 0$ satisfies $\min_m F_\lambda(c)_m/c_m \geq 1$, then $c/\min_m c_m$ is a feasible solution of $L_k^{NT}(\lambda)$.*

Proof. $c \leq F_\lambda(c)$ is the displayed condition; homogeneity preserves it under rescaling; the rescaled vector has min $= 1$ and is finite, so (L0) holds with $C^{\max} = \max c/\min c$; (L1)–(L4) hold with $\bar{c}$ as auxiliaries by construction. □

This is the easy (verification) direction of the Collatz–Wielandt formula for monotone homogeneous maps [3]; the useful empirical fact, standard in nonlinear Perron–Frobenius theory, is that normalized power iteration $c \leftarrow F_\lambda(c)/\|F_\lambda(c)\|_\infty$ converges to vectors achieving $\min_m F(c)_m/c_m$ close to the spectral value, so that whenever $\lambda < \lambda_k$ the iteration eventually exhibits a certificate in the sense of Lemma 3.1. We stress that no claim in this paper depends on convergence theory: the iteration is only a search heuristic, and Lemma 3.1 is checked directly on its output.

Three practical devices matter at scale. (i) Each sweep is two gather operations and one three-way minimum over arrays of length 3^{k-1}; $k = 17$ sweeps take under a second vectorized. (ii) Bisection on λ re-uses the certificate of the last feasible λ as a warm start. (iii) For the largest k we skip bisection entirely: by Proposition 2.2 the certificate from level $k-1$ lifts to level k, and a single targeted run at a chosen λ either certifies it or not — any certificate is final, so nothing is lost by aiming directly at a target exponent.

4 Exact certification

Let the claimed exponent be $\gamma = p/q \in \mathbb{Q}$ and $\lambda = 2^{p/q}$, so $\log_2 \lambda = p/q$ exactly. The three coefficients of F_λ are then q-th roots of explicit rationals:

$$\lambda^{-2} = \left(\tfrac{1}{2^{2p}}\right)^{1/q}, \qquad \lambda^{\alpha-2} = \left(\tfrac{3^p}{2^{2p}}\right)^{1/q}, \qquad \lambda^{\alpha-1} = \left(\tfrac{3^p}{2^p}\right)^{1/q}.$$

Lemma 4.1. *Let $u/v \leq (A/B)^{1/q}$ with u, v, A, B positive integers, which is equivalent to the integer inequality $u^q B \leq v^q A$. If the system obtained from $L_k^{NT}(2^{p/q})$ by replacing each coefficient with such a rational lower bound admits a solution, then so does $L_k^{NT}(2^{p/q})$ itself.*

Proof. All coefficients occur positively on right-hand sides of $\leq$-constraints; decreasing them only shrinks the right-hand sides. □

The verifier therefore proceeds as follows. Rational lower bounds with $v = 2^{64}$ are computed for the three coefficients and certified by the integer comparison of Lemma 4.1 (computed once; the integers involved have a few hundred thousand bits for our $q \leq 2500$). The floating-point vector c found in Section 3 is scaled by 2^{48}, floored to an integer vector C, and clamped below at 2^{48} (so $c = C/2^{48} \geq 1$). Each constraint is then checked as a single integer inequality; for example (L3) becomes

$$C_m \cdot v_2 v_C \;\leq\; u_2 v_C \cdot C_{4m} \;+\; u_C v_2 \cdot \min\bigl(C_u, C_{u+3^{k-1}}, C_{u+2\cdot 3^{k-1}}\bigr),$$

with (u_2, v_2) and (u_C, v_C) the certified bounds for λ^{-2} and $\lambda^{\alpha-1}$. A run that checks all 3^{k-1} constraints yields, by Lemmas 3.1 and 4.1 and Theorem 2.1, the bound $\pi_a(x) \geq x^{p/q}$. No floating-point number survives into the verified statement.

5 Results

Validation. Bisection with the iteration of Section 3 reproduces the published table of [5, Table 2] for $2 \leq k \leq 11$ at all seven printed decimals; in particular the record entry $\lambda_{11} = 1.7922310$ is matched exactly.

New levels. Beyond $k = 11$:

k	classes 3^{k-1}	λ certified	$\gamma = \log_2 \lambda$	exact certificate p/q
12	177,147	1.8064235	0.8531361	$853/1000 = 0.853$
13	531,441	1.8188238	0.8630058	$863/1000 = 0.863$
14	1,594,323	1.8307723	0.8724524	$2181/2500 = 0.8724$
15	4,782,969	1.8419683	0.8812483	$2203/2500 = 0.8812$
16	14,348,907	≥ 1.8506089	≥ 0.888	$111/125 = 0.888$
17	43,046,721	≥ 1.8596099	≥ 0.895	$179/200 = 0.895$

For $k \leq 15$ the third column is the certified end of a bisection (the true λ_k exceeds it by less than 2×10^{-7}); for $k = 16, 17$ it is a single targeted certificate as described in Section 3, and the true λ_k is somewhat larger. In every row the final column was verified by the exact integer procedure of Section 4; at $k = 17$ this comprises 43,046,721 verified inequalities.

Proof of Theorem 1.1. The exact certificate at $k = 17$ exhibits a feasible solution of $L_{17}^{NT}(2^{179/200})$ by Lemmas 3.1 and 4.1. Apply Theorem 2.1. $\square$

6 Machine verification of the record

The exponent 0.895 of Section 5 rests on Theorem 2.2 of [5] and an exact integer check of the certificate in Python. This section reports a fully machine-verified exponent above the 2003 record: everything from the dynamics of T to the final inequality is a Lean 4 proof, with the certificate checked by the Lean kernel itself.

The obstacle: the advanced term

Krasikov's difference inequalities for the class infima $\phi_k^m(y) = \inf\{\pi_a^*(2^y a) : a \equiv m\ (3^k)\}$ read ([5], Prop. 2.1; $\alpha = \log_2 3$)

$$
\begin{aligned}
m \equiv 2\ (9): &\quad \phi^m(y) \geq \phi^{4m}(y-2) + \bar{\phi}^{(4m-2)/3}(y+\alpha-2),\\
m \equiv 5\ (9): &\quad \phi^m(y) \geq \phi^{4m}(y-2),\\
m \equiv 8\ (9): &\quad \phi^m(y) \geq \phi^{4m}(y-2) + \bar{\phi}^{(2m-1)/3}(y+\alpha-1),
\end{aligned}
$$

where $\bar{\phi}$ is the minimum over the three lifts of a class of modulus 3^{k-1}. The last term is *advanced*: the child $(2a-1)/3$ is smaller than a, so its tree is counted at a larger relative scale $y+\alpha-1$. A feasible solution c of the linear program with coefficients $\lambda^{-2}, \lambda^{\alpha-2}, \lambda^{\alpha-1}$ yields $\phi^m(y) \geq \Delta c_m \lambda^y$ (their Theorem 2.2), but a straightforward induction on y cannot invoke the hypothesis at $y+\alpha-1 > y$. Krasikov and Lagarias prove Theorem 2.2 through a back-substitution procedure that eliminates advanced terms (their §3–5), with a deletion rule justified by monotonicity and termination proved via Kőnig's lemma; the derived system has nested minimizations of depth > 226 already at $k = 5$. Truncating the advanced term instead, i.e. using monotonicity to retard it below y, is what Applegate–Lagarias did in 1995 and what the earlier formalization `Krasikov50.lean` does; we computed that the truncated system's exponent saturates near 0.66 at every level, while the exact system reaches 0.849 at $k = 12$: the advanced term is worth about 0.19 in the exponent, and cannot be dispensed with.

A root induction instead of elimination

We work on the 1/50 grid of `Krasikov50.lean`: caps $\mathrm{cap}(t) = 2^{\lfloor t/50 \rfloor} r[t \bmod 50]$ with a fixed integer table $r \approx 10^4 \cdot 2^{i/50}$, rate $\mu = p/q$ per grid step, so that $\lambda = \mu^{50}$ and every certificate condition is an integer inequality. The three inequalities become, with offsets in grid steps,

$$\phi(m, t+100) \;\ge\; \phi(4m, t) \;+\; \begin{cases} \bar{\phi}\big(\frac{4m-2}{3}, t+79\big) & m \equiv 2 \ (9), \\ 0 & m \equiv 5 \ (9), \\ \bar{\phi}\big(\frac{2m-1}{3}, t+129\big) & m \equiv 8 \ (9), \end{cases}$$

the offsets $79 = 100 - 21$ and $129 = 100 + 29$ being the grid roundings of $2 + (\alpha - 2)$ and $2 + (\alpha - 1)$ (kernel-checked table conditions $4\,\mathrm{cap}(t) \le 3\,\mathrm{cap}(t+21)$ and $2\,\mathrm{cap}(t+29) \le 3\,\mathrm{cap}(t)$; the earlier file used 99 in place of 129, which is the truncation).

The key observation is that the advanced term can be handled by a plain strong induction provided one inducts over *roots* a rather than classes, because root size is a second well-founded quantity that decreases exactly where the scale increases. Define

$$M(t, a) \;=\; 10\,t + \lfloor \log_2 a^{498} \rfloor.$$

Along the three branches, with $b = (2a-1)/3$:

$$\begin{aligned} (4a,\ t-100):&\quad 10t - 1000 + \lfloor \log_2 (4a)^{498} \rfloor = 10t + \lfloor \log_2 a^{498} \rfloor + 996 - 1000, \\ (2b,\ t-21):&\quad \lfloor \log_2 (2b)^{498} \rfloor \le \lfloor \log_2 a^{498} \rfloor + 207 \text{ (as } 3 \cdot 2b \le 4a), \quad -210 + 207, \\ (b,\ t+29):&\quad \lfloor \log_2 b^{498} \rfloor \le \lfloor \log_2 a^{498} \rfloor - 291 \text{ (as } 3b \le 2a), \quad +290 - 291, \end{aligned}$$

so M drops by 4, 3 and 1 respectively; the two inequalities on logarithms both reduce to the single numeric fact $3^{498} > 2^{789}$ ($498 \log_2 3 = 789.3$). The induction then goes through verbatim: the certificate inequality at the class of a, the three per-root difference inequalities (which are the internals of the class inequalities, with the child's class now known exactly, so the minimum over lifts is replaced by the certificate value at that class), and the induction hypothesis at the three children. Two features of the grid are essential. The exact advanced shift $\alpha - 1 = \log_2(3/2)$ would make the measure degenerate — the conditions on the exponent N in $\lfloor \log_2 a^N \rfloor$ become $2N < 100\beta$ and $N \log_2(3/2) > 29.248\,\beta$, i.e. $N < 50\beta$ and $N > 50\beta$ — which is precisely why Krasikov–Lagarias needed elimination; the grid offset $29 < 50 \log_2(3/2) = 29.248$ opens the window $49.58\beta < N < 50\beta$ (we use $\beta = 10$, $N = 498$), at a cost of 1/50 of a doubling on that branch. And inducting over roots removes the minimum over lifts from the induction and the need for nonemptiness witnesses for the 3^{k-1} classes.

Theorem 6.1 (`G50.growth_root`)**.** *Let $k \ge 2$, $1 \le q \le p$, and $c : \mathbb{N} \to \mathbb{N}$ with $1 \le c_m \le C_{\max}$ on the classes $m \equiv 2\ (3)$ of modulus 3^k, satisfying*

$$\begin{aligned} m \equiv 5\ (9) :&\quad c_m p^{100} \le c_{4m} q^{100}, \\ m \equiv 2\ (9) :&\quad c_m p^{100} \le c_{4m} q^{100} + \bar{c}_{(4m-2)/3}\, p^{79} q^{21}, \\ m \equiv 8\ (9) :&\quad c_m p^{100} q^{29} \le c_{4m} q^{129} + \bar{c}_{(2m-1)/3}\, p^{129}. \end{aligned}$$

Then for every root $a \ge 3$, $a \equiv 2\ (3)$, whose orbit reaches 8, and every t,

$$c_{a \bmod 3^k}\, p^t q^{100} \;\le\; \pi_a^*\big(\mathrm{cap}(t)\, a\big) \cdot \big(C_{\max}\, q^t p^{100}\big).$$

The certificate and the kernel check

At $k = 12$ (177,147 classes) the grid system admits $\mu = 101182/100000$, i.e. $\lambda = \mu^{50} = 1.7995$ and $\gamma = 50 \log_2 \mu = 0.8476$, with an integer certificate of 32-bit entries found by power iteration on the monotone map of Section 3 (exponents $-100, -21, +29$ in place of $-2, \alpha - 2, \alpha - 1$) and checked exactly in Python. In Lean the certificate is 81 packed natural-number literals of 2187 entries each, read by kernel-accelerated shift and mask; the per-class condition is a Boolean function built from `Nat.ble`, evaluated over each shard by a divide-and-conquer range checker of logarithmic depth with a once-proved soundness lemma, and discharged by `decide +kernel` — the kernel's own reduction, no `native_decide`, no trusted compiler — in about fifteen minutes and 18 GB. The final theorem:

Theorem 6.2 (`K12.density_bound`)**.** *For every $y \ge 0$,*

$$\begin{aligned} &554204 \cdot 101182^{50y} \cdot 100000^{100} \\ &\qquad \le\; \#\{n \le 80000 \cdot 2^y : \textit{the orbit of } n \textit{ reaches } 1\} \cdot \big(16777216 \cdot 100000^{50y} \cdot 101182^{100}\big), \end{aligned}$$

i.e. $\pi_1(x) = \Omega(x^\gamma)$ with $\gamma = 50 \log_2(101182/100000) = 0.84763\ldots$.

Axioms: `propext`, `Classical.choice`, `Quot.sound` only; no `sorry`. This is the first machine-verified exponent above 0.84 — the published record since 2003 — and, we believe, the first machine verification of any result in the Krasikov–Lagarias program at full strength. The grid rounding costs 0.0055 against the exact linear program (0.8531 at $k = 12$); a finer grid recovers it. The same files instantiate at any level: the kernel time grows a little faster than linearly in the number of classes, so $k = 13$ (0.8569) is about an hour and $k = 14$ (0.870) a few hours; the informal 0.895 at $k = 17$ is the same system at a level the kernel has not yet been asked to check. The $k = 13$ instance has also been run: `K13.density_bound` (files `KL13Data`, `KL13Check0--2`, `KL13`; 531,441 classes in 243 shards, 2 h 15 min of kernel time) gives $\gamma = 50 \log_2(101195/100000) = 0.8569\ldots$, with the analogous displayed inequality ($387326 \cdot 101195^{50y} \cdot 100000^{100} \le \#\{\cdots\} \cdot (16777216 \cdot 100000^{50y} \cdot 101195^{100})$).

Trust base, stated precisely

What is verified: the definitions of T, the capped backward tree and its count, the three difference inequalities for every root, the growth theorem, the certificate conditions for all 177,147 classes, and the final density inequality. What is not verified and need not be: how the certificate was found (power iteration in floating point, then rounded). What is imported:

nothing beyond Lean's kernel and Mathlib. The $k = 12$ modules were additionally replayed through an independent instance of the kernel by `lean4checker` (built for the same toolchain), which re-evaluates every shard proof from the stored declarations and reports no discrepancy. What is claimed: exactly the displayed inequality, for every y, with no "sufficiently large".

7 Discussion

The marginal exponent gains per level, $+0.0114, +0.0099, +0.0094, +0.0088$ across $k = 11, \ldots, 15$, decay slowly; whether $\lambda_k \to 2$ (equivalently $\pi_a(x) \geq x^{1-\epsilon}$ for all ϵ) remains the fundamental open question about this method [5, §6], and the data is consistent with it. Each additional level costs a factor of 3 in memory and time; $k = 18$ (129M classes) is within reach of a workstation, and $k \approx 20$ of a GPU implementation. The bottleneck is not the certification (linear scan) but the iteration count near the feasibility threshold, which warm starts mitigate.

Three remarks on reliability. First, everything rests on Theorem 2.1, which is published and peer-reviewed; we add no asymptotic analysis of our own, and the new content of this paper is checkable by a referee in exact arithmetic from the certificate files. Second, each certificate was checked by *two independently written verifiers*: the original, and a clean-room re-implementation using only the language's standard library (no shared code, no numerical libraries), with the constraint system re-derived from [5, §2], a different rational bounding scheme for the coefficients (denominator 10^{21} versus 2^{64}, binary search versus unit adjustment), and explicit assertions of the congruence bookkeeping; both verifiers also detect a single corrupted entry (negative control). Third, the verification procedure is small (under a hundred lines) and is itself a candidate for formalization: the certificate check is a finite integer computation of precisely the kind proof assistants discharge by reflection, and formalizing Theorem 2.1 — an induction over difference inequalities — would make the exponent fully machine-verified.

Machine-verified exponents, and the critical constant

That program is underway in the companion repository [8], which contains complete Lean 4 formalizations of integer-grid weakenings of the difference-inequality method at two strengths, both proved from the dynamics upward (backward-tree combinatorics, Krasikov's inequalities, kernel-checked certificates; axioms: Lean's standard `propext`, `Classical.choice`, `Quot.sound` only — no `sorry`, no extra axioms).

1. **Unit grid** (file `Krasikov.lean`): caps $2^y \cdot a$, an 81-class certificate at modulus 3^5 with rational rate $137/100$, giving the machine-checked bound $\pi_1(x) = \Omega(x^{0.4543})$ (theorem `density_lower_bound`).

2. $1/50$ **grid** (file `Krasikov50.lean`): caps follow the interleaved-geometric sequence $\mathrm{cap}(t) = 2^{\lfloor t/50 \rfloor} r[t \bmod 50]$ for a fixed 50-entry integer table $r \approx 10^4 \cdot 2^{i/50}$, so that the per-step rate is the *rational* $\mu = 1261/1250$ and every certificate condition is a pure integer inequality. The two odd-branch terms of the difference inequalities retard by only $21/50$ resp. $1/50$ of a doubling (the kernel-checked table conditions; the margin is $2^{21/50} = 1.3382\ldots > 4/3$). A 2187-class certificate at modulus 3^8, decided by the kernel in chunks, together with per-class witness roots $a = m + 6561q$ ($q \in \{0, 1\}$) whose Terras orbits reach 8 in at most 146 kernel-evaluated steps, yields (theorem `K8.density_bound`, since retired from

the repository tree in favour of the results of Section 6; its definitional core survives as `Grid50.lean`):

$$\pi_1(x) \;=\; \Omega(x^\gamma)\,, \qquad \gamma \;=\; 50\log_2\frac{1261}{1250} \;=\; 0.632009\ldots$$

The second exponent crosses a meaningful line: it exceeds $\log 2/\log 3 = 0.630929\ldots$, the critical odd-step density against which every quantity of this problem is measured (it is the density a divergent orbit would have to sustain forever, and the constant governing the cycle Diophantine bounds). The comparison of the two numbers is numerical rather than structural — they measure different things — but as a milestone it is sharp: the machine-verified count of integers reaching 1 now provably grows faster than $x^{\log 2/\log 3}$. The exponent 0.632 exceeds every bound published before Applegate–Lagarias [1]; the formal pipeline has since been scaled past the 2003 record itself (Section 6).

Reproducibility

The search engine, the one-shot driver, both exact verifiers, the certificate export format (raw little-endian integers with a hashed sidecar), and instructions are in the repository accompanying this paper (`analysis/kl_exponent.py`, `kl_oneshot.py`, `kl_certificate.py`, `kl_export_certificate.py`, `kl_verify_independent.py`); all runs are deterministic. Finding the $k = 17$ certificate takes under an hour on a laptop; verifying it exactly takes minutes per verifier.

Acknowledgments

The computations, the certification scheme, and this text were produced in collaboration with Claude (Anthropic). We thank the authors of [5] for a paper written so completely that improving its constant required no new mathematics — only twenty-three years of Moore's law and one change of algorithm.

References

[1] D. Applegate and J. C. Lagarias, Density bounds for the $3x + 1$ problem. II. Krasikov inequalities, *Math. Comp.* **64** (1995), 427–438.

[2] R. E. Crandall, On the "$3x + 1$" problem, *Math. Comp.* **32** (1978), 1281–1292.

[3] S. Gaubert and J. Gunawardena, The Perron–Frobenius theorem for homogeneous, monotone functions, *Trans. Amer. Math. Soc.* **356** (2004), 4931–4950.

[4] I. Krasikov, How many numbers satisfy the $3x + 1$ conjecture?, *Internat. J. Math. Math. Sci.* **12** (1989), 791–796.

[5] I. Krasikov and J. C. Lagarias, Bounds for the $3x+1$ problem using difference inequalities, *Acta Arithmetica* **109** (2003), 237–258.

[6] J. C. Lagarias (ed.), *The Ultimate Challenge: The* $3x+1$ *Problem*, American Mathematical Society, 2010.

[7] J. C. Lagarias, The $3x + 1$ problem: an overview, arXiv:2111.02635 (2021).

[8] M. Sharpe, The Collatz conjecture at the critical line: machine-verified no-go, density, and cycle theorems, companion paper and repository, 2026.

[9] T. Tao, Almost all orbits of the Collatz map attain almost bounded values, *Forum of Mathematics, Pi* **10** (2022), e12.

[10] G. J. Wirsching, *The Dynamical System Generated by the* $3n+1$ *Function*, Lecture Notes in Mathematics 1681, Springer, 1998.